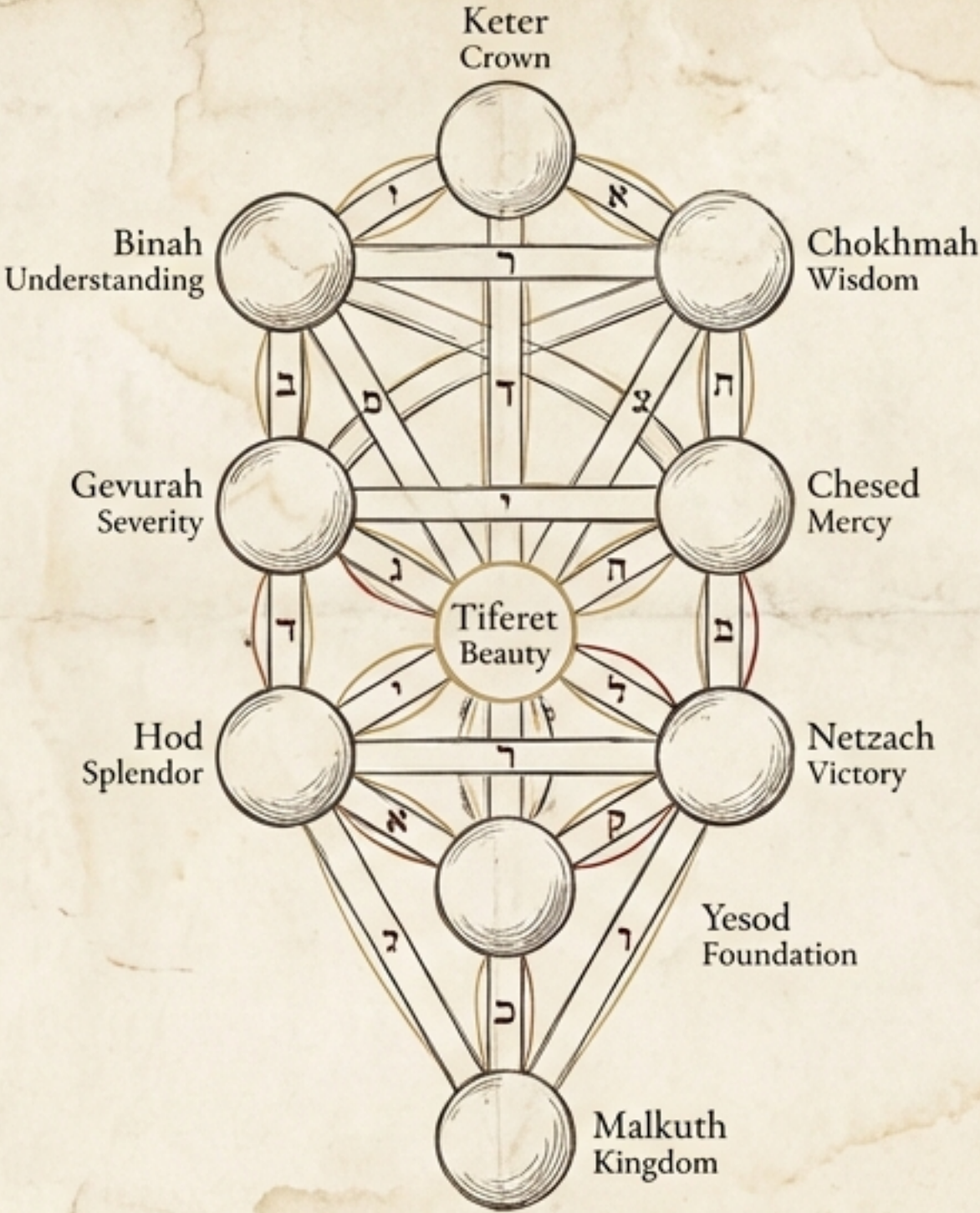


TET~CRAFT: The Two Worlds of a Simulation

An exploration of the dialogue between metaphysical blueprints and mechanical reality in the creation of an emergent universe.



```
@jit(fastmath=True, cache=True)
def world_update_physics_jit(state, entities, forces, constraints, dt):
    var state = ...
    entities = entities.entities.count
    var noutats = ...
    entities = ...
    for i in range(entities.count):
        forces[i] = forces[i]
        forces[i] = forces.metand(i)met()
        entities.position[i] += entities.position[leas[0] - 1]
        ...
        entities.position[i] += entities.position[i] += state
    for in i state update:
        state = entities.count
        forces = forces.countcount()
        entities.position[state + forces, metand[0], 0)
        entities.position[i] += state:metandree()
    return ...
```

High-performance JIT compilation

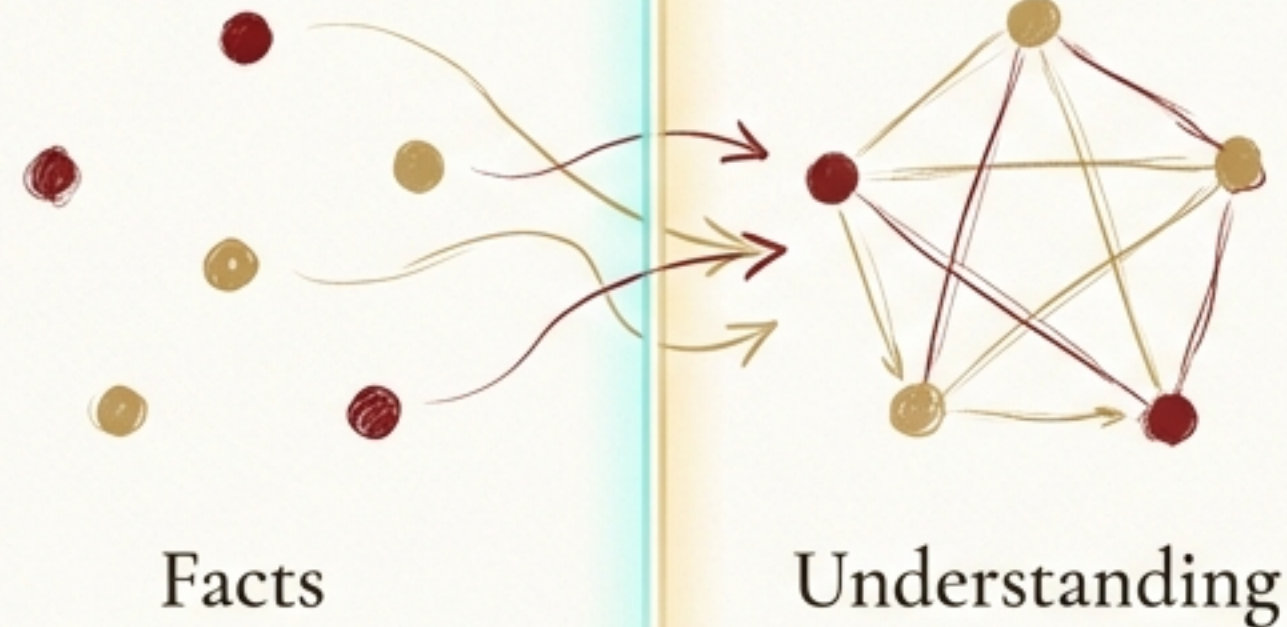
Physics integration step

Entity state update

What if a Universe Could Be Simulated from First Principles?

Each TET represents a quantum fact that can combine with others to form understanding.

“This model assumes the player as God, the prime observer missing from quantum theory. Pressing space creates a new fact(TET) to be misunderstood.”



- 4 faces (White, Black, Red, Cyan)
- Battery energy (0-1)
- Magnetic polarity (+/-)



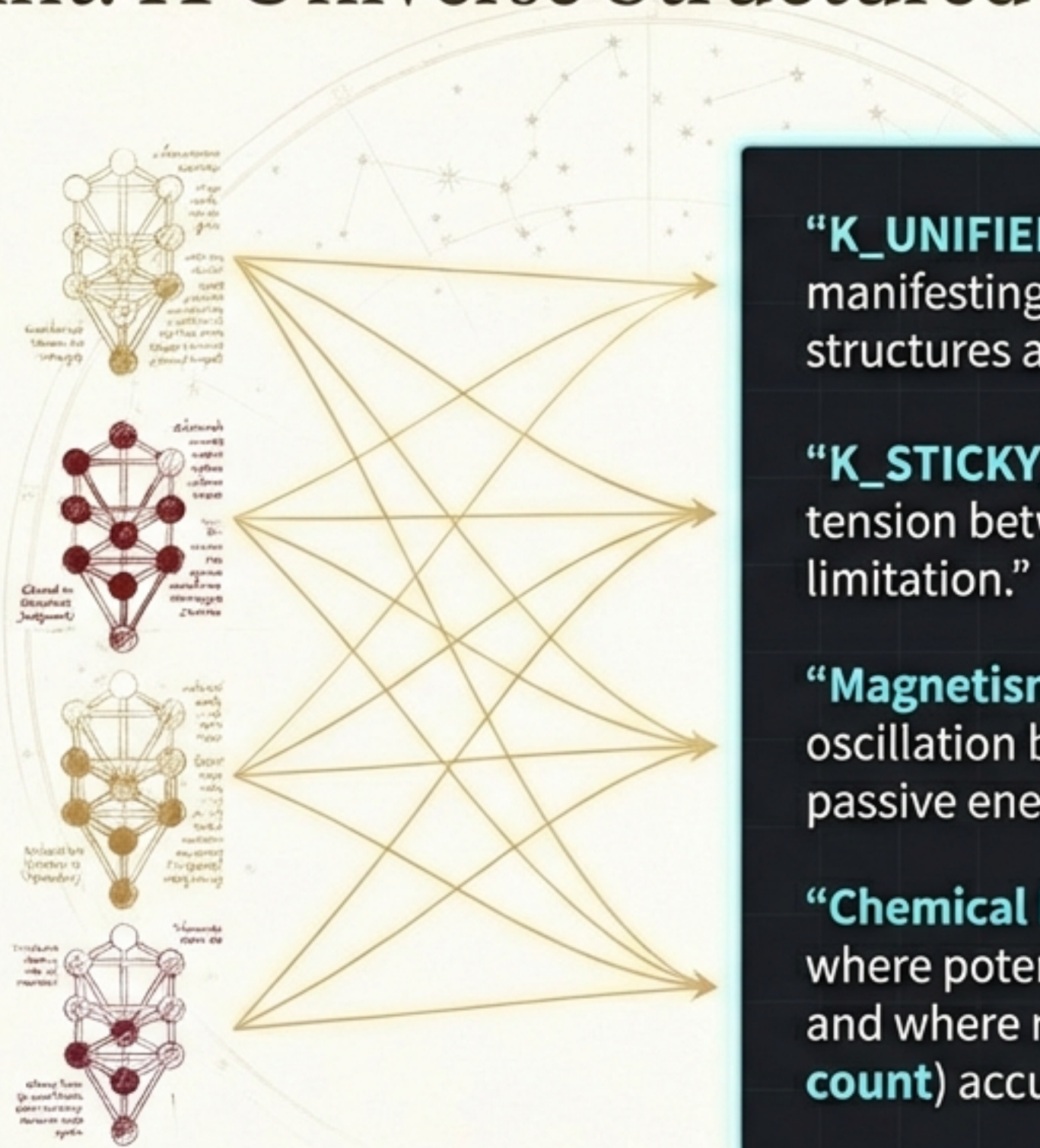
The Blueprint: A Universe Structured as the Tree of Life

Keter to Malkhut
(Crown to Kingdom)

Chesed to Gevurah
(Mercy to Judgment)

Netzach to Hod
(Victory to Splendor)

Yesod (Foundation)

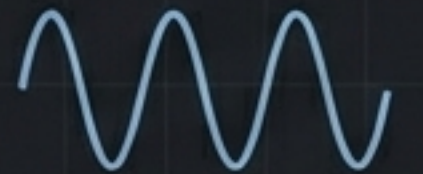


“K_UNIFIED_FORCE”: “God’s will manifesting as the law that structures all existence.”

“K_STICKY_PULL”: “The creative tension between expansion and limitation.”

“Magnetism”: “The eternal oscillation between active and passive energies.”

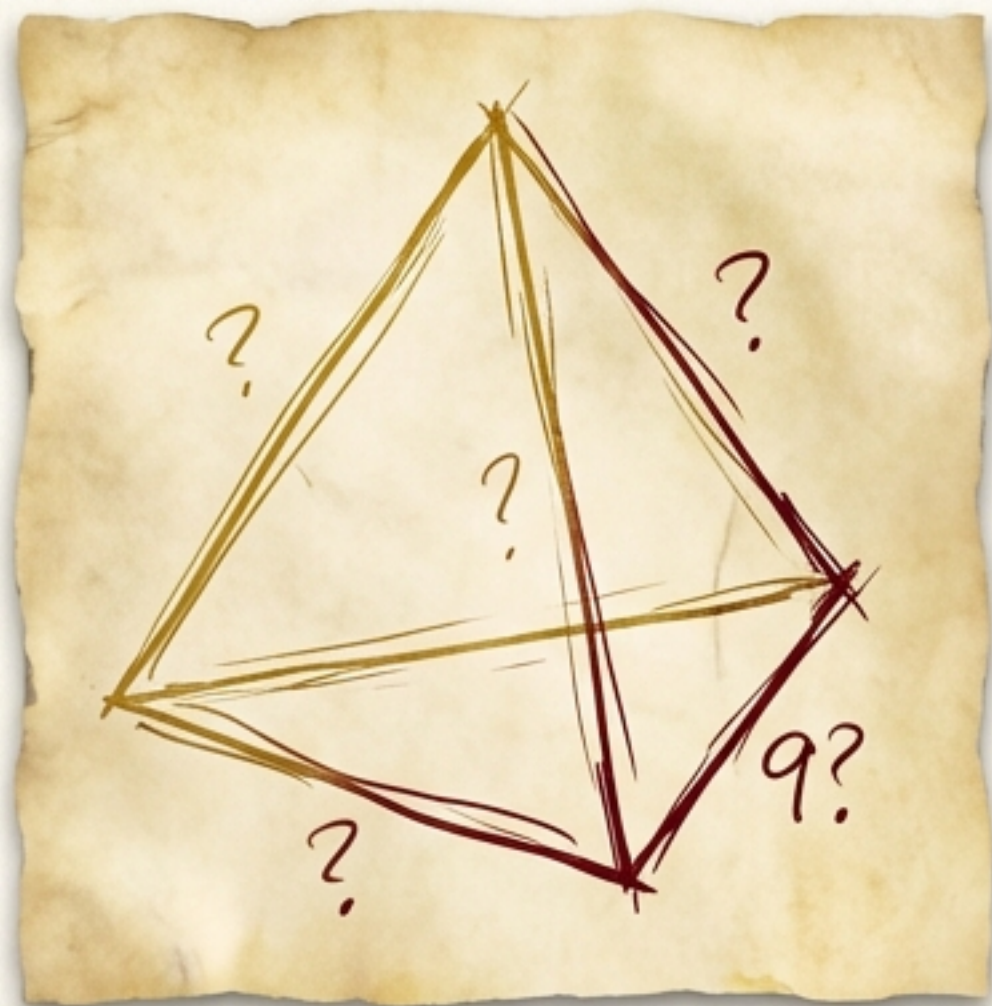
“Chemical Reactions”: “The realm where potential forms actualize... and where memory (**synthesis_count**) accumulates.”



“The 4D Past Projection as Ain Sof (the Infinite)...
the boundless light that precedes and envelops the structured tree.”

But a Universe Needs Laws. The First is Geometry.

*“A tetrahedron has 4 sides...
likewise it has 9? edges...”*



The Python physics engine defines a tetrahedron with mathematical precision.

```
EDGES_NP = np.array([(i, j) for i in range(4)
                        for j in range(i+1, 4)])
```

The `world_update_physics_jit` function iterates through exactly 6 edges. The geometry is fixed.



The Second Law: Geometry is Destiny

Explains the Interface Chemistry system, where the physical alignment of tetrahedral faces defines the identity of matter.

The Interface Chemistry Table

Interface	Element	Symbol	Attribute
Cyan ↔ Red	Hydrogen	H	Abundance
White ↔ White	Carbon	C	Balance
White ↔ Black	Nitrogen	N	Structure
Red ↔ Red	Iron	Fe	Magnetic
Black ↔ Black	Helium	He	Purity
Red ↔ White	Oxygen	O	Catalyst

```
INTERFACE_CHEMISTRY = {  
    (2, 3): "H", # Red-Cyan  
    (0, 0): "C", # White-White  
    (0, 1): "N", # White-Black  
    (2, 2): "Fe", # Red-Red  
    # ...  
}
```

“Elements are defined by WHICH faces touch. Labeling a TET ‘Gold’ does nothing. You must physically connect specific faces to transmute matter.”

The Inhabitant: A Mind of Pure Observation

“To understand Bot Intelligence in this system, imagine a room full of heat-seeking drones in total darkness.”

Advanced Mechanics

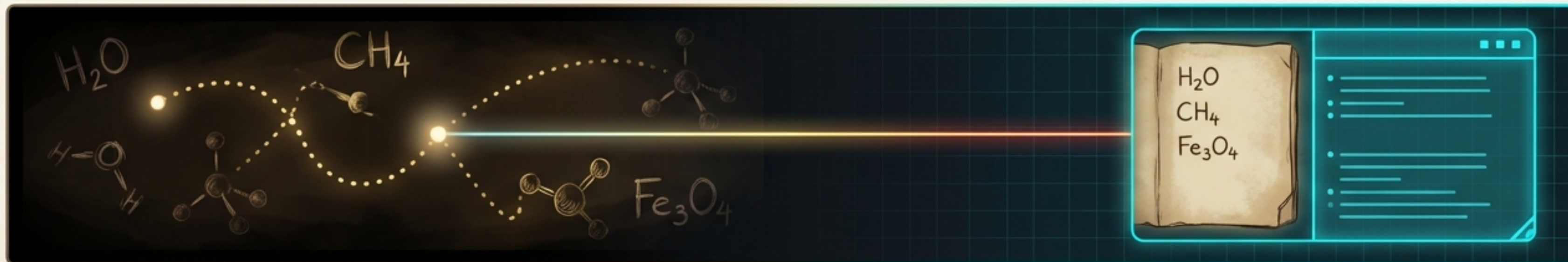
“The infrared sensors that allow the drones to find the ‘warmth’ of energy-rich molecules.”

BotMind

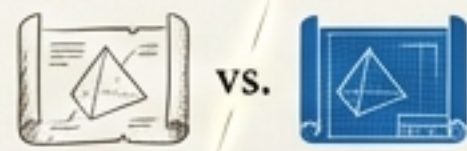
“The internal logbook where the drone records what it finds... slowly turning a room of random collisions into a documented story of chemistry.”

System Functions

- Automates exploration in headless mode.
- Spawns new TETs with chemical labels.
- Generates chemical thoughts using molecule names (`get_thought_symbol`).
- Tracks synthesis champions.



Divergence Case Study: Geometry and Transcendence



Blueprint vs. Building

Divergence 1: The Shape of Reality

The Vision - Blueprint

"The notes speculate on the structure:

"a tetrahedron has ... 9? edges".



The Machine - Building

"The Python physics engine (EDGES_NP array) correctly iterates through 6 edges."

```
np.array([(i, j) for i in range(4)
          for j in range(i+1, 4)])
```



Divergence 2: The Cost of Enlightenment

The Vision - Blueprint

'The 'Cosmic Ladder' summary table lists the requirement for Era 7: Transcendence as 200 pts.'

7. Transcendence	200 pts	"Beyond matter"
------------------	---------	-----------------

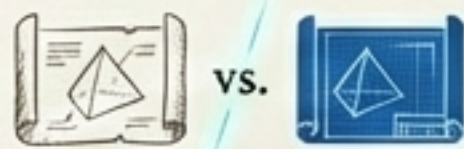
The Machine - Building

'The Python `TechTree` class defines the threshold as 777 pts. Descriptive text confirms this.'

```
('Transcendence', 777, 'Beyond matter')
```



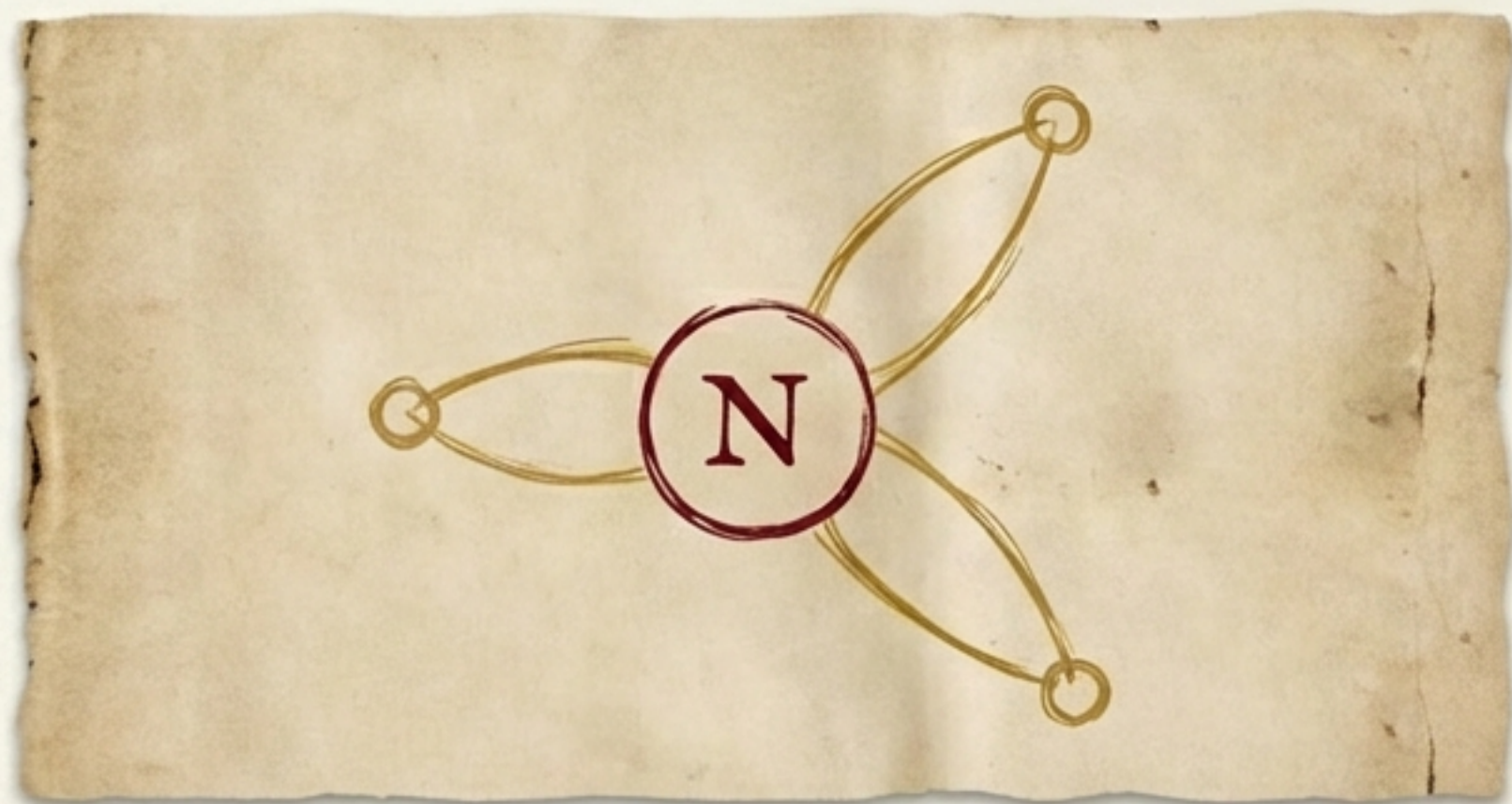
Divergence Case Study: The Potential of Nitrogen



Blueprint vs. Building

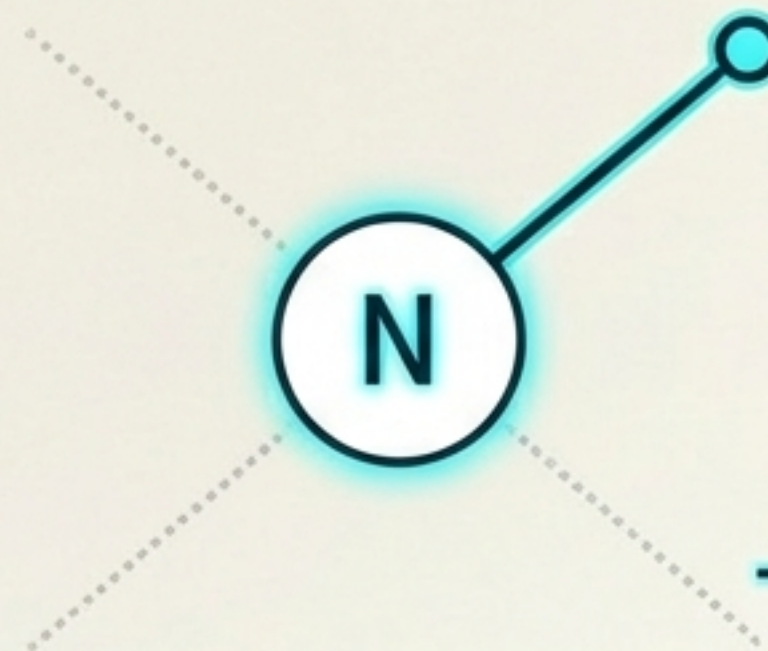
The Core Conflict The Vision - Blueprint

“The ‘Chemical Kabbalah’ theory describes Nitrogen as having a valency of 3, allowing it to support structural complexity.”



The Coded Capacity The Machine - Building

“In the Python configuration for `BOND\_CAPACITY`, Nitrogen is not explicitly listed, meaning it falls under the default capacity of 1.”

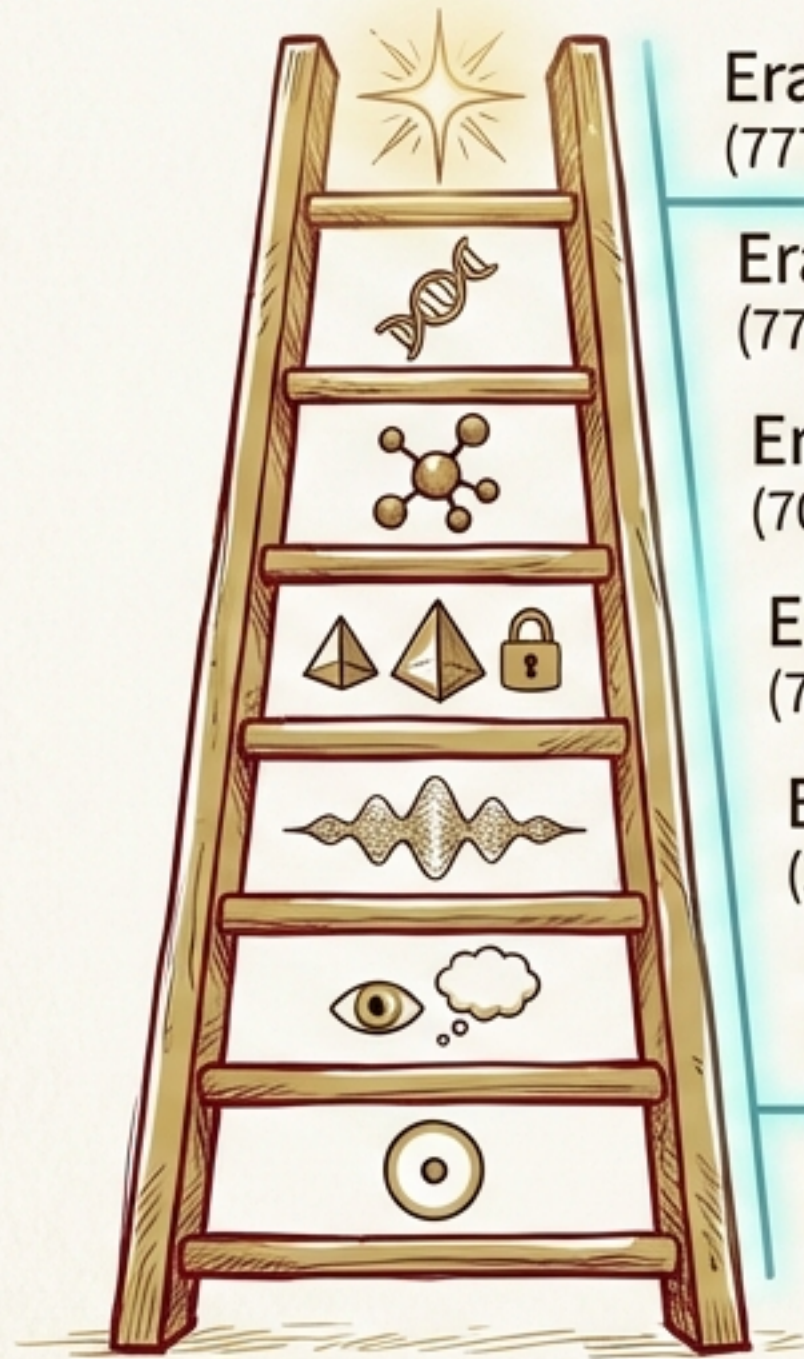


```
BOND_CAPACITY = {  
    'C': 4,  
    'Ce': 4,  
    'Fe': 2,  
    'default': 1  
}
```

“The architect’s ambitious ‘3-bond’ designs are currently held together by a single ‘default’ bolt.”

The Journey to Transcendence: A Ladder Through Two Worlds

Introduces the “Cosmic Ladder” as the player’s progression system.
Progress is not bought, but achieved by aligning the machine’s state with metaphysical goals.
“Progress is not measured by time, but by **Complexity** and **Entropy reduction**.”



Era 7: Transcendence
(777 pts)

Era 6: Life
(777 pts)

Era 5: Chemistry
(70 pts + 7 Molecules)

Era 4: Condensation
(7 pts + 0.3 Phi)

Era 3: Fluctuation
(3 Atoms + 0.1 PSI)

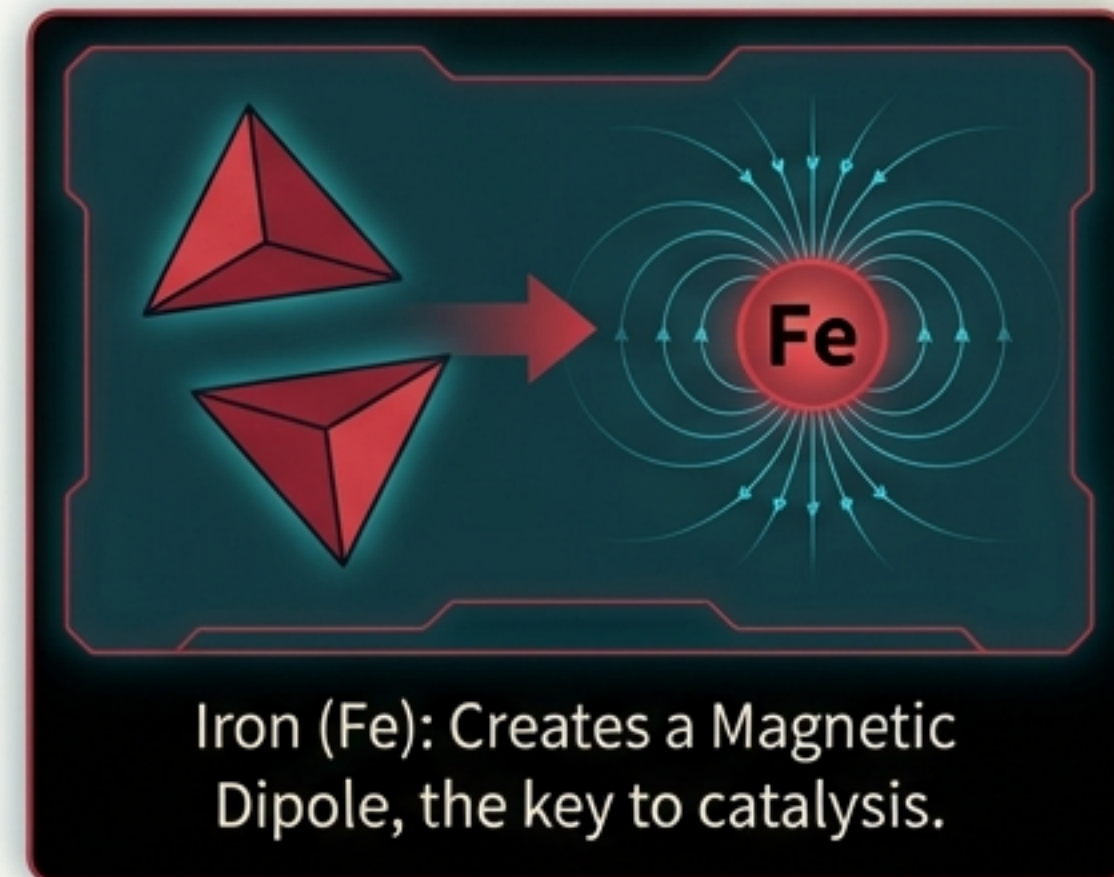
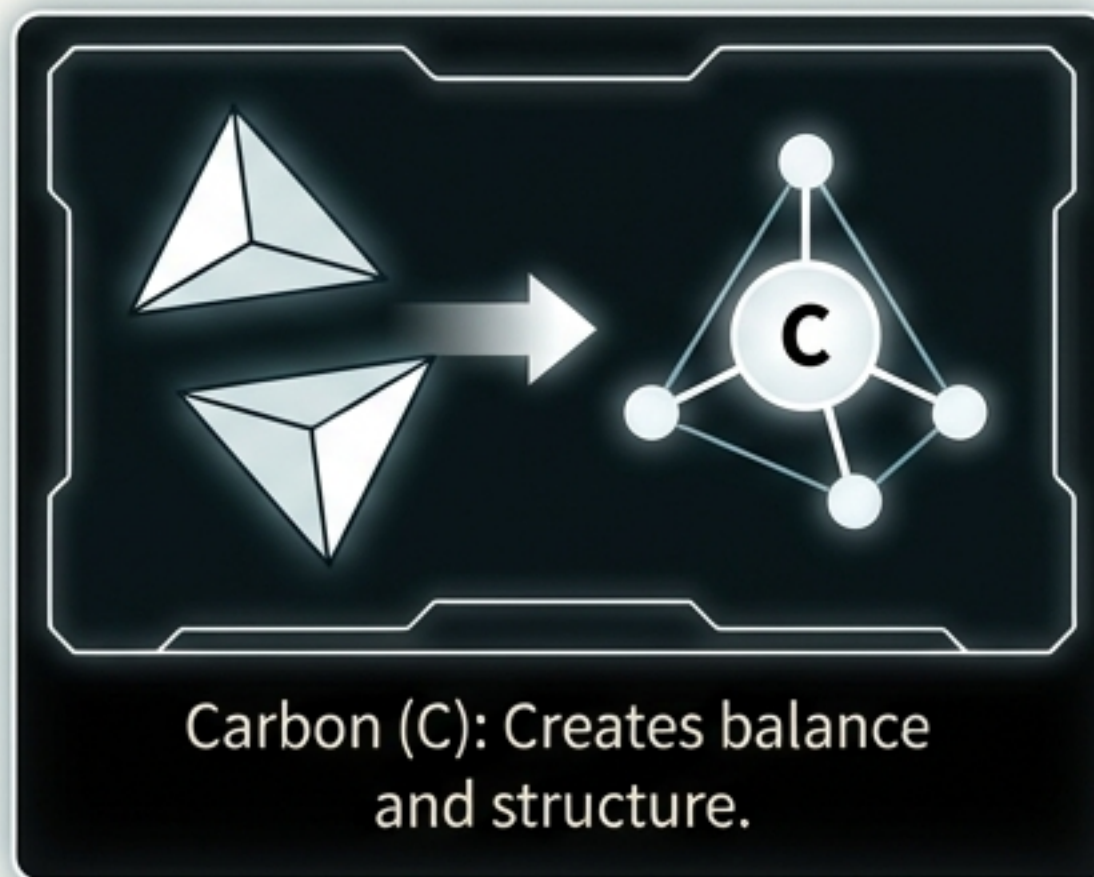
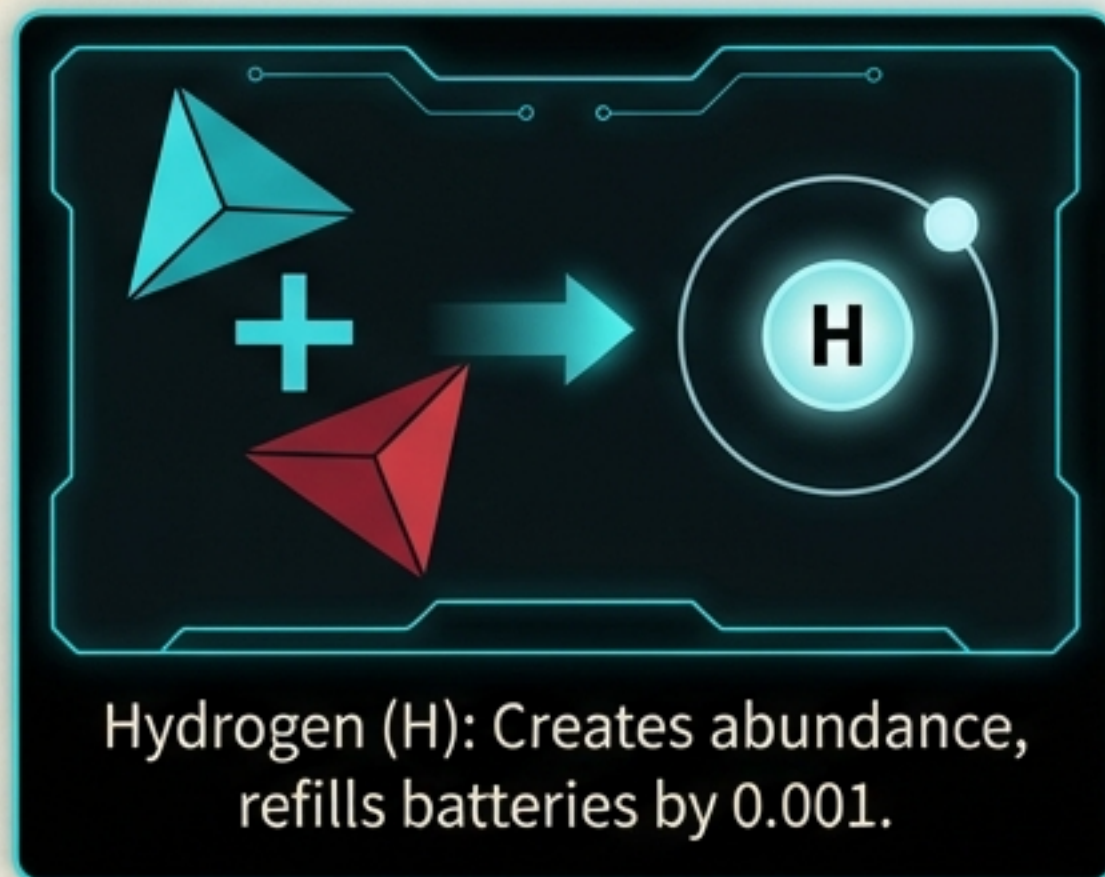
Era 2: Mind
(3 TETs)

Era 1: Void
(0 pts)

“You cannot ‘buy’ upgrades.
You cannot ‘cheat’ by renaming things.
Geometry is Destiny.”

Speaking the Language of the Machine

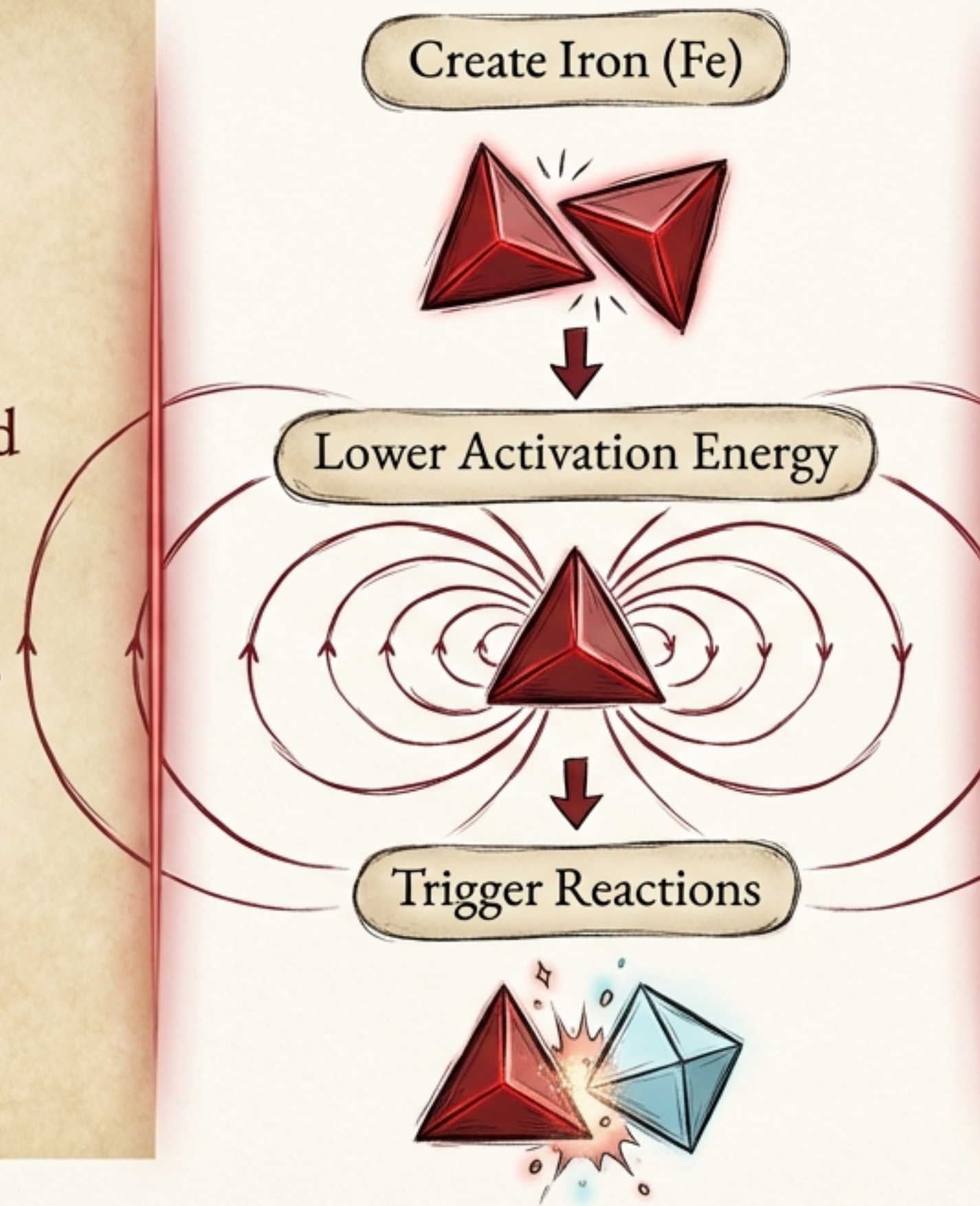
A deep dive into the Interface Chemistry system as the player's primary creative tool.



By physically locking specific faces, the player transmutes matter, creating the building blocks required to advance. This is the synthesis of player intent (the vision) and the system's rules (the machine).

The Catalyst Strategy: Wielding Judgment to Spark Life

“From the Kabbalistic tradition, the color Red corresponds to *Gevurah* (Strength or Judgment). It signifies intensity, power, and severity.”



“In TET~CRAFT, the Red-Red interface creates Iron, a catalyst that wields magnetic power to force reactions. The player uses an act of Judgment (*Gevurah*) to overcome entropy and create Life.”

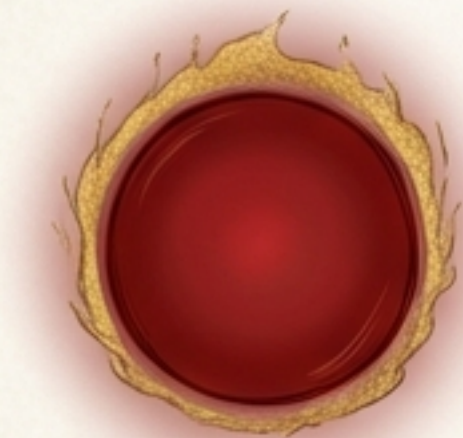
A Deeper Look: The Chemical Kabbalah

Section 1: The Sefirot of Color



Corresponds to Chesed (Kindness, Mercy).

In the simulation, the White ↔ White interface creates Carbon, the foundation of balanced structures.



Corresponds to Gevurah (Judgment, Strength).

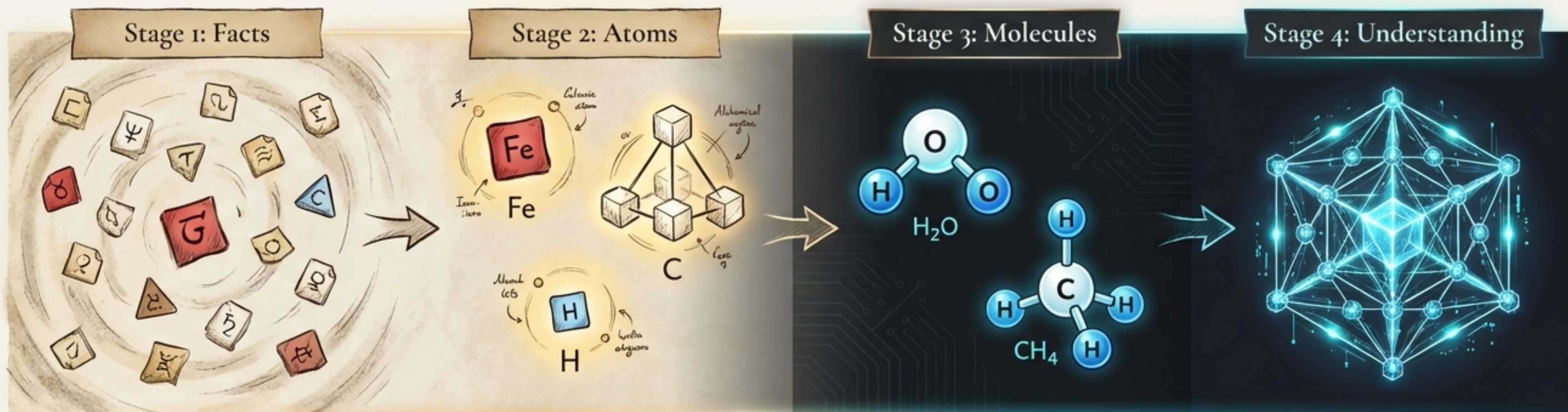
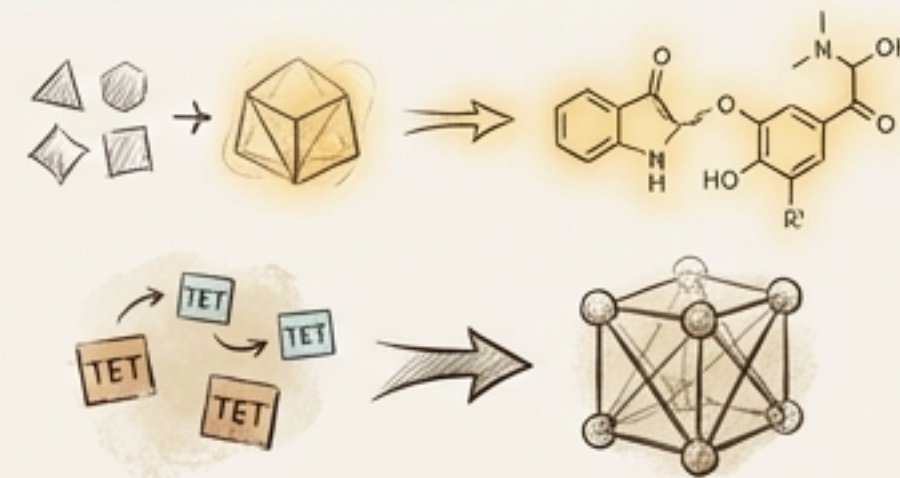
In the simulation, the Red ↔ Red interface creates Iron, the magnetic catalyst of power.

Section 2: The Full Tree of Forces

Force/Concept	Sefirotic Correspondence	Function
K_UNIFIED_FORCE	Keter to Malkhut	Gravitational pull toward center.
K_STICKY_PULL	Chesed to Gevurah	Attraction between TETs.
Magnetism	Netzach to Hod	Positive/negative polarity.
Corner Desires	Tiferet	Red↔Cyan attraction.
Chemical Reactions	Yesod	Synthesis/decomposition processes.
The TETs Themselves	Malkhut	A crystallized 'fact' from the void.

The End Goal: Simulating Understanding

- Emergent Chemistry: “To create a system where complex, real-world chemistry arises spontaneously from a simple set of underlying geometric rules.”
- Formation of Understanding: “To simulate how disconnected facts (TETs) can combine through interaction and observation to form diologetti from sun observation to form coheirent structures of understanding (molecules).”



“The player (Keter) peers through Da’at (camera) to witness Malkhut (TETs) ascending and descending the tree... the alpha and omega collapsing into one another, the tree circling back to its root.”

The Fourth Temple

(A Communication Protocol)

Version: v5.3D

License: Apache 2.0

Metaphysics: Supplied by cenezer & Jesus

Misunderstood Physics: Gemini 2.5

Black Hole Revision: Gemini 3.0

Chemistry: Deepseek (~v3.3)

Aesthetics: Opus 4.5

Thermodynamics Whitepaper: GPT-4o

Entire Project Vibe: cenezer (20/12/20-Current)

"A kleinverse of tetrahedrons that bond into atoms, react, and evolve into complex molecules... a temple still under construction."